Lecture notes on Thermal Physics

Based on the syllabus according to National Education Policy (NEP) 2020, India.

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Abstract: This e-content has been prepared for the B.Sc. second semester students in accordance with NEP syllabus. The manuscript is based on the lectures I have delivered in class during academic session 2021-2022. Along with basic concepts of thermal physics, problems have also been discussed. Thermal radiation and kinetic theory of gases is not in the scope of this write-up. Moreover, multiple answer type questions are given as an exercise to the students. I feel that this e-content will bring students more closer to the subject and help them to think beyond. References followed for prepartion, are given at the end. The e-content will not be used for commercial use of any kinds. Thanks.

Keywords: Thermodynamics, phase transition, entropy, Carnot's Engine.

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1 Basic introduction to thermodynamics

1.1 Thermodynamical systems

A system is defined as a definite quantity of matter i.e. solid, liquid or gas bounded by some closed surface so that it can be investigated.

1. **Homogeneous system:** A system whose properties are independent of the position and are uniform throughout the system, is called a homogeneous system.

2. **Hetrogeneous system:** A system whose properties are non-uniform and vary within the system, is called a hetrogeneous system.

Classification of systems

On the basis of surroundings and exchanging properties, the systems are of three following types

1. **Open System:** An open system is a system which can exchange energy and matter with the surroundings.

2. **Closed System:** An closed system is a system which can exchange only energy and not matter with the surroundings.

3. **Isolated System:** An open system is a system which can neither exchange energy and nor matter with the surroundings. Here, anything outside the system is termes as surroundings.

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1.2 Equilibrium

The physical systems can be found in three types of equilibria

1. **Mechanical Equilibrium:** For a system to be in Mechanical Equilibrium, there should be no unbalanced forces acting on the system i.e., there should be no microscopic movement inside the system.

2. **Thermal Equilibrium:** For a system to be in Thermal Equilibrium, there should be no temperature difference within the systems or between the system and surroundings.

3. **Chemical Equilibrium:** For a system to be in chemical equilibrium, there should be no chemical reactions between the diferent paths of the systems or between the system and surroundings.

1.3 Variables

Every thermodynamical process is completely defined in terms of co-ordinates called *thermodynamical variables*. Pressure (p), volume (V) and temperature (T) are the examples of thermodynamical variables. Let u be parameter specifying the state of a homogeneous system. Consider that the system is divided into two parts by a partition. Then $u_1 \& u_2$

will be the values of this parameter for the two subsystems, then y is said to be *extensive* if

$$u = u_1 + u_2 \quad \text{e.g., Volume} \tag{1.1}$$

and it is said to be *intensive* if

$$u = u_1 = u_2 \quad \text{e.g., Temperature} \tag{1.2}$$

1. **Extensive Variables:** Variables that are proportional to the mass and the number of particles of the system are called extensive variables e.g., volume, energy, mass, length and etc.

2. **Intensive Variables:** A specific variable corresponding to any extensive variable which is independent of mass is called intensive varialbe e.g., pressure, temperature, C_P, density, melting point, boiling point and etc.

1.4 Equation of state

Equation of a state equation of state is an equation which determines the state of a system and its condition at that instant, this state is determined by certain observable parameters like temperature, pressure, volume etc. the general form of the equation of state is

$$f(p, V, T) = 0 \quad \text{(general or parametric form)} \tag{1.3}$$

1. For an ideal gas

$$p\ V = n\ R\ T \tag{1.4}$$

2. For a real gas, e.g., Van der waals gas

$$\left(p + \frac{a}{V_2}\right)(V - b) = R\ T \tag{1.5}$$

1.5 Thermodynamical Processes

1. **Reversible process:** A reversible process is the one which can be retraced in the opposite direction by changing the external conditions infinitely slow. This means if heat is absorbed by the system in the direct process, the same quantity of heat will be given out in the opposite process, there is no net loss of energy in the reversible process. The major conditions for any process to be reversible are

   (a) The process must be quasi-static (*Dheere-Dheere*) i.e., it should proceed at a infinitely slow rate.

   (b) There should be zero loss of energy due to conduction or radiation.

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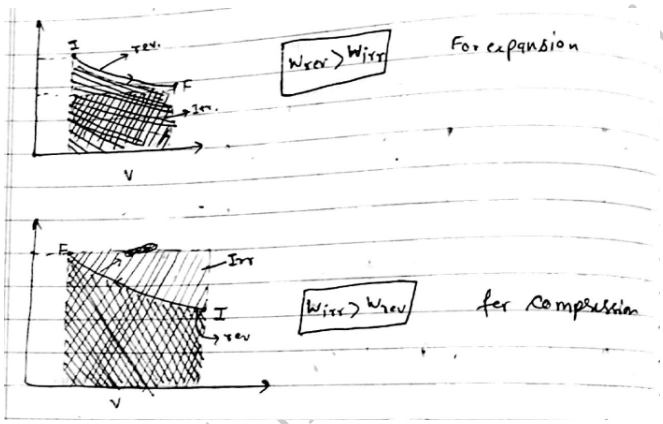
2. **Irreversible:** Vice-versa.

   Workdone in reversible and irriversible processes: For expansion

$$W_{reversible} > W_{irriversible} \tag{1.6}$$

   For compression

$$W_{reversible} < W_{irriversible} \tag{1.7}$$



3. **Cyclic process:** A process which does not retrace its path, is named as a cyclic process.

4. **Isothermal Process:** It is a process which takes place at a constant temperature i.e., the system undergoes a change in such a way that it's temperature remains constant.

$$T = constant$$

5. **Adiabatic Process:** An adiabatic process is a process in which no heat enters the system or leaves the system i.e.,

$$\delta Q = 0$$

   In this process, the *Entropy (S)* of the system remains constant. According to II-law of thermodymanics (which we will study in the later section)

$$\begin{aligned} \delta Q &= T\ dS \\ \because \delta Q &= 0 \\ \therefore dS &= 0 \\ \Rightarrow S &= constant \end{aligned}$$

6. **Isobaric process:** It is a process in which the pressure (p) remains constant.

$$\begin{aligned} p &= constant \\ \Rightarrow dp &= 0 \end{aligned}$$

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7. **Ischoric process:** It is a process in which volume (V) remains constant.

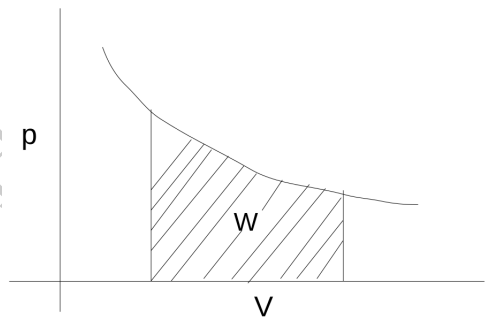
$$V = constant$$
$$\Rightarrow dV = 0$$

A note on $\delta(.)$ and $d(.)$ To write the differentails of heat and work done we'll use the symbol δ i.e., δQ & δW beacause these quantities are imperfect differentials, they cannot be represented as a function of the state moreover they are path functions (depend on the path taken). For internal energy and entropy, we'll use the symbol dU and dS respectively because these are perfect (exact) differentials and can be represented as a function of state, these functions are path independent.

1.6 Work done

work done for a process in carrying the system from state 1 to state 2 is found by computing the area under the $p - V$ curve.

$$W = \int_{\text{state1}}^{\text{state2}} p \, dV \tag{1.8}$$



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Work done for an isothermal process

let us consider an ideal gas which undergoes expansion from volume V_1 to V_2, then the work done at constant temperature is calculated as

$$W = \int_{V_1}^{V_2} p \, dV$$
$$\because PV = n \; RT \quad \text{(for an ideal gas)}$$
$$\Rightarrow p = \frac{n \; RT}{V}$$
$$\therefore W = \int_{V_1}^{V_2} \frac{nRT}{V} \, dV$$

$$\boxed{\Rightarrow W = n \; RT \; ln\left(\frac{V_2}{V_1}\right)}$$

Work done for an isobaric process

Isobaric process means a process at a constant pressure. Let us consider an ideal gas which undergoes expansion from volume V_1 to V_2, then the work done at constant pressure is calculated as

$$\begin{aligned} W &= \int_{V_1}^{V_2} p\ dV \\ \because p &= constant \\ \Rightarrow W &= p\ \int_{V_1}^{V_2} dV \\ \Rightarrow W &= p\ (V_2 - V_1) \end{aligned} \tag{1.9}$$

Work done for an isochoric process

A thermodynamical process at constant volume, is called a isochoric process.

$$\begin{aligned} W &= \int_{V_1}^{V_2} p\ dV \\ \because V &= constant \\ \Rightarrow W &= 0 \end{aligned} \tag{1.10}$$

Q: What is the sign convention followed for work?

When a system expands against an external pressure, $dV = +ve\ \Rightarrow W = +ve$ and we say the work is done by the gas. When a system is compressed againts the gas pressure, then $dV = -ve\ \Rightarrow W = -ve$ and we say the work is done on the gas.

1.7 Internal Energy

The sum of total internal kinetic energy and potential energy of all molecules of a system constitute the internal energy of the system. If it increses in the system, a positive sign is used. If it decreases in the system, a negative sign is used.

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2 Zeroth and first law of thermodynamics

2.1 Zeroth law of thermodynamics

If two systems are separately in thermal equilibrium with the third system, then they will also be in thermal equilibrium with one another.

2.2 First law of thermodynamics and its applications

The first law is essentially a statement of law of conservation of energy for thermodynamical systems. It states the equivalance of heat and energy. Let dU be the change in the internal energy of the system, when δQ amount of heat is added to it such that δW is the work done by it and dN is the change in the number of the particles, then the first law can be written as

$$dU = \delta Q \ - \ \delta W \ + \ \mu dN \tag{2.1}$$

where μ is the chemical potential of the system. Above equation is the differential form of first law of thermodynamics. For a non diffusively interacting system $dN = 0$ (i.e., # particles = constant)

$$dU = \delta Q \ - \ \delta W \tag{2.2}$$

Significances of first law

- The first law establishes heat as a form of energy.
- It introduces the concept of internal energy and provides a method for determining change in internal energy.
- It is the statement of law of conservation of energy.

Limitations

- It does not specify how much of heat energy can be converted into work.
- It also does not specify the condition under which a system can use its heat energy to produce a supply of mechanical work.

2.3 Applications of first law of thermodynamics

Heat capacities

$$\begin{aligned} \because \delta Q &= dU \ + \ \delta W \\ &= dU \ + \ pdV \end{aligned} \tag{2.3}$$

Defining heat capacity at constant volume as

$$C_V = \left(\frac{\delta Q}{\delta T}\right)_V \tag{2.4}$$

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from equation 2.3, at constant volume $\delta Q = dU$

$$\therefore C_V = \left(\frac{\partial U}{\partial T}\right)_V \tag{2.5}$$

and defining heat capacity at constant pressure as

$$C_p = \left(\frac{\delta Q}{\delta T}\right)_p \tag{2.6}$$

Let U be the function of T & V, the total derivative dU is given by

$$\begin{aligned} U &\equiv U\ (T, V) \\ dU &= \left(\frac{\partial U}{\partial T}\right)_V dT\ +\ \left(\frac{\partial U}{\partial V}\right)_T dV \end{aligned} \tag{2.7}$$

sunstituting dU in equation 2.3, we get

$$\delta Q = \left(\frac{\partial U}{\partial T}\right)_V dT\ +\ \left[p\ +\ \left(\frac{\partial U}{\partial V}\right)_T\right]\ dV \tag{2.8}$$

dividing above equation by dT at constant pressure,

$$\left(\frac{\delta Q}{\delta T}\right)_p = \left(\frac{\partial U}{\partial T}\right)_V + \left[p\ +\ \left(\frac{\partial U}{\partial V}\right)_T\right]\ \left(\frac{\partial V}{\partial T}\right)_p$$

$$\boxed{\Rightarrow C_p - C_V = \left[p\ +\ \left(\tfrac{\partial U}{\partial V}\right)_T\right]\left(\tfrac{\partial V}{\partial T}\right)_p}$$

students are encouraged to remember above relation.

Q:For an ideal gas prove that $C_p\ - C_V\ =\ R$ (also known as meyer's law)

Since there is only kinetic energy for an ideal gas as there is no interaction between molecules of an ideal gas, $\left(\frac{\partial U}{\partial V}\right)_T = 0$

from last equation

$$C_p\ -\ C_V = p\ \left(\frac{\partial V}{\partial T}\right)_p \tag{2.9}$$

Now considering the equation of state for one mole of an ideal gas

$$\begin{aligned} pV &= RT \\ \therefore \left(\frac{\partial V}{\partial T}\right)_p &= \frac{R}{p} \\ \Rightarrow C_p\ - C_V\ &=\ R \end{aligned} \tag{2.10}$$

Q: For a Van Der Waals (a real) gas prove that $C_p\ - C_V\ =\ R\left[1\ +\ \frac{2a(V-b)^2}{RTV^3}\right]$

Hint: start with equation of state for a real gas i.e., $\left(p\ +\ \frac{a}{V^2}\right)\ (V\ - b)\ =\ RT$, show that $\left(\frac{\partial U)}{\partial V}\right)_T = \frac{a}{V^2}$.

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Q: Show that the first law of thermodynamics can be written as

$$\delta Q \;=\; C_V dT \;+\; pdV$$

Solution: From the definition of C_V

$$C_V = \left(\frac{\delta Q}{\delta T}\right)_V \;=\; \left(\frac{\partial U}{\partial T}\right)_V$$

$$\Rightarrow\; dU \;=\; C_v\; dT$$

According to I-law, $\quad \delta Q = dU \;+\; \delta W$

$$\Rightarrow\; \delta Q = C_V\; dT \;+\; pdV$$

Q: Show that for an ideal gas, the first law of thermodynamics can be written as

$$\delta Q \;=\frac{C_p pdV}{R} \;+\; \frac{C_V V dp}{R}$$

Solution: From the first law of thermodynamics

$$\delta Q = C_V\; dT \;+\; pdV \tag{2.11}$$

for an ideal gas

$$pV = RT$$

$$\Rightarrow\; pdV + Vdp = RdT$$

$$\Rightarrow\; pdV = RdT - Vdp \tag{2.12}$$

substituting (2.12) in (2.11),

$$\begin{aligned}\delta Q &= C_V dt \;+\; RdT \;-\; Vdp\\ &= (C_V + R)\; dT \;-\; Vdp\\ &= C_p\; dT \;-\; Vdp\end{aligned} \tag{2.13}$$

also from (2.12)

$$dT = \frac{1}{R}\,(pdV + Vdp)$$

$$\therefore \delta Q = \frac{C_p}{R}\,(pdV + Vdp) \;-\; Vdp$$

$$= \frac{C_p\; p\; dV}{R} \;+\; \frac{C_p\; V\; dp}{R} \;-\; \frac{R\; V\; dp}{R}$$

$$\boxed{\Rightarrow\; \delta Q \;= \frac{C_p pdV}{R} \;+\; \frac{C_V V dp}{R}}$$

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Poisson's equation for an adiabatic process

The differential form of first law,

$$\delta Q = dU \ + \ pdV$$
$$\Rightarrow \ dU = -pdV \tag{2.14}$$

If a gas is permitted to expand adiabatically it does work at the cost of its internal energy i.e., the internal energy of the gas is decreases and hence, its temperature falls during adiabatic expansion. Similarly, adiabatic compression always produces an increase in internal energy and hence an increase in temperature. Equation (2.14) can be written as,

$$C_V \ dT \ + \ pdV = 0$$

since, for an ideal gas, $pV = RT$

$$C_V dT \ + \ \frac{RT}{V} \ dV = 0 \tag{2.15}$$

dividing above equation by $C_V \ T$,

$$\frac{1}{T} \ dT + \ \frac{R}{C_V} \ \frac{1}{V} dV = 0 \tag{2.16}$$

integrating,

$$lnT \ + \ ln\left(V^{R/C_V}\right) = ln\kappa$$
$$\Rightarrow T \ V^{\frac{C_p - C_V}{C_V}} = constant$$

since $\frac{C_p}{C_V} = \gamma$, therefore we get

$$\boxed{T \ V^{\gamma-1} = constant}$$

using equation of state for an ideal gas, one can deduce

$$p \ V^{\gamma} = constant \tag{2.17}$$
$$T^{\gamma} \ p^{1-\gamma} = constant \tag{2.18}$$

Cyclic process

When a thermodynamical process takes place via closed path i.e. initial state and final states are same, is called a cyclic process. In easy words, consider a priceess from point a to b. Now if be return from b to a, but via different path, the process a to b to a will be a cyclic process. From first law of thermodynamics

$$\delta Q = dU \ + \ \delta W$$

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for a closed path

$$\oint \delta Q = \oint dU \ + \ \oint \delta W \tag{2.19}$$

since internal energy U does not depend on the path taken, but only depends on the state, the change in internal energy will be zero

$$dU = 0$$

therefore

$$\boxed{\oint \delta Q = \oint \delta W}$$

Q: Show that the ratio of adiabatic to isothermal elasticity is equal to γ.
Solution: The expressions for isothermal and diabatic elasticit are given by,

$$E_T = -V \left(\frac{\partial p}{\partial V}\right)_T \tag{2.20}$$

$$E_S = -V \left(\frac{\partial p}{\partial V}\right)_S \tag{2.21}$$

<u>For isothermal process</u>

$$pV = RT$$
$$\Rightarrow pdV \ + \ Vdp = 0 \quad \text{(partial diff.)}$$
$$\Rightarrow \left(\frac{\partial p}{\partial V}\right)_p = -\frac{p}{V}$$

therefore, isothermal elasticity,

$$E_T = p \tag{2.22}$$

<u>For adiabatic process</u>

$$pV^\gamma = constant$$
$$\Rightarrow \left(\frac{\partial p}{\partial V}\right)_S = -\frac{p}{V}\ \gamma$$

therefore, adiabatic elasticity,

$$E_T = \gamma\ p \tag{2.23}$$

the ratio of adiabatic to isothermal elasticity will be

$$\boxed{\frac{E_S}{E_T} = \gamma}$$

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Q: Starting from first law of thermodynamics, prove that

$$\left(\frac{\partial U}{\partial V}\right)_T = \frac{C_p - C_V}{V\alpha_p} - p$$

Solution: from first law, we have already derived

$$C_p - C_V = \left[p + \left(\frac{\partial U}{\partial V}\right)_T\right] \left(\frac{\partial V}{\partial T}\right)_p \tag{2.24}$$

by definition, coefficient of expansion

$$\alpha_p = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_p \tag{2.25}$$

therefore

$$C_p - C_V = \left[p + \left(\frac{\partial U}{\partial V}\right)_T\right] V\,\alpha_p$$

$$\Rightarrow \left(\frac{\partial U}{\partial V}\right)_T = \frac{C_p - C_V}{V\alpha_p} - p \tag{2.26}$$

Q: Starting from first law of thermodynamics, prove that

$$\left(\frac{\partial U}{\partial T}\right)_p = C_p - V\alpha_p \tag{2.27}$$

Solution: Wrtiting U as a function of p & T

$$U \equiv U(p, T)$$

the total derivative of U will be

$$dU = \left(\frac{\partial U}{\partial p}\right)_T dp \;+\; \left(\frac{\partial U}{\partial T}\right)_p dT$$

from first law of thermodynamics,

$$\delta Q = dU \;+\; pdV$$

substituting dU

$$\delta Q = \left(\frac{\partial U}{\partial p}\right)_T dp \;+\; \left(\frac{\partial U}{\partial T}\right)_p dT \;+\; pdV$$

for an isobaric process,

$$\left(\frac{\partial U}{\partial T}\right)_p = \left(\frac{\partial U}{\partial T}\right)_p \;+\; p\,\left(\frac{\partial V}{\partial T}\right)_p \qquad (\because dp = 0)$$

$$\Rightarrow C_p = \left(\frac{\partial U}{\partial T}\right)_p \;+\; pV\;\alpha_p$$

$$\Rightarrow \left(\frac{\partial U}{\partial T}\right)_p = C_p - V\alpha_p \tag{2.28}$$

Q: Starting from first law of thermodynamics, prove that

$$\left(\frac{\partial U}{\partial p}\right)_T = \frac{pV}{E_T} \;-\; \frac{C_p - C_V}{E_T\alpha_p}$$

Solution: Try it yourself.

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Work done during an adiabatic process

Consider an adiabatic process during which a gas undergoes adiabatic expansion form state a to state B. From first law of thermodynamics

$$dU = \delta Q \ - \ \delta W$$

for an adiabatic process $\delta Q = 0$

$$\begin{aligned} dU &= -\delta W \\ \Rightarrow C_V \ dT &= - \ \delta W \end{aligned} \tag{2.29}$$

$$\boxed{\delta W \ = \ - \ C_V \ \Delta T}$$

in other form,

$$\boxed{W \ = \ \frac{P_2V_2 - P_1V_1}{1-\gamma}}$$

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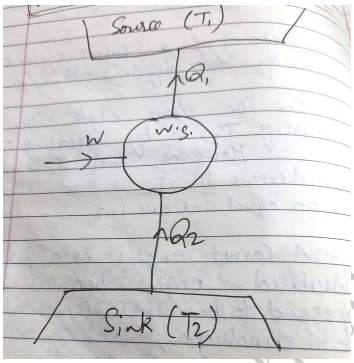
3 Heat Engines

3.1 Carnot's Engine

A heat engine is a machine that can convert heat into work. To be an useful device, the engine must be operated continuously in a cycle.



Essentials

1. Must operate between teo reservoires
2. Working substance
3. Efficiency

$$\eta = \frac{\text{useful work done}}{\text{amount of heat absorbed from dource}} = \frac{W}{Q_1}$$

$$\boxed{\eta = 1 \; - \; \frac{Q_2}{Q_1}} \; \eta < 1$$

ideal heat engine

Components

1. Cylinder: it has perfectly non-conduction walls, perfectly conducting base and having a non-conducting piston which moves without friction. The cylinder has ideal gas as a working substance.
2. Heat insulating stand: a perfectly non-conducting platform acts as stand for adiabatic process.
3. Source: heat reservoire at temperature T_1 having infinite thermal capacity.
4. Sink: it a reservoire at maintained at temperature T_2 to which the heat engine can reject any amount of heat. it too has infinite thermal capacity. In order to obtain a continuous supply of work, the working substance in cornot's engine i.e., gas is subjected to cycle of *quasi-static* opertaions, known as the *Carnot's cycle*.

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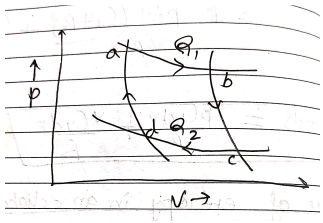

From state a to state b

From styate a to b the working substance in the cylinder is brought in the thermal contact with the heat reservoire at T_1, the gas undergoes reversible isothermal expansion and absorbs heat Q_1 to attain state b. Let W_1 be the work done during this process.

$$\begin{aligned} W_1 &= \int p dV \\ &= \int_a^b \frac{nRT_1}{V}\, dV \\ &= nRT_1\, ln\left(\frac{V_b}{V_a}\right) \end{aligned} \tag{3.1}$$

From state b to state c

During this process, the gas is thermally isolated and it is allowed to undergo reversible adiabatic expansion. During this process, the temperature of gas falls from T_1 to T_2. Let W_2 be the work done in going from state b to c.

$$\begin{aligned} W_2 &= -\ C_V\ \Delta T \\ &= C_V\ (T_1 - T_2) \end{aligned} \tag{3.2}$$

From state c to state d

The cylinder with working substance is placed in thermal contact with the cold reservoire at T_2 and allowed to undergo reversible isothermal compression. During this process amount of heat Q_2 is given to the sink.

$$W_3 = nRT_2\ ln\left(\frac{V_d}{V_c}\right) \tag{3.3}$$

From state d to state a

The gas is again thermally isolated and compressed under reversible adiabatic conditions till the original state as restored. Work is done on the gas and is equal to

$$\begin{aligned} W_4 &= -\ C_V\ \Delta T \\ &= C_V\ (T_2 - T_1) \end{aligned} \tag{3.4}$$

Therefore the total work done is given by

$$\begin{aligned} W &= W_1\ +\ W_2\ +\ W_3\ +\ W_4 \\ &= nRT_1\ ln\left(\frac{V_b}{V_a}\right)\ +\ nRT_2\ ln\left(\frac{V_d}{V_c}\right) \end{aligned} \tag{3.5}$$

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Since the points b and c lie on the same adiabat, therefore

$$T_1\ V_b^{\gamma-1} = T_2\ V_c^{\gamma-1}$$
$$\Rightarrow \frac{T_1}{T_2} = \left(\frac{V_c}{V_b}\right)^{\gamma-1} \tag{3.6}$$

similarly for points a and d

$$\frac{T_1}{T_2} = \left(\frac{V_d}{V_a}\right)^{\gamma-1} \tag{3.7}$$

comparing (3.6) and (3.7)

$$\frac{V_b}{V_a} = \frac{V_c}{V_d}$$

substituting this relation in total work done formula, we have

$$\boxed{W = nR\ (T_1 - T_2)\ ln\left(\frac{V_b}{V_a}\right)}$$

$$\because \delta Q = C_V\ dT + \delta W$$

for a to b i.e., isothermal process $\delta Q = \delta W$, therefore

$$Q_1 = W_1$$
$$= nR\ T_1\ ln\left(\frac{V_b}{V_a}\right) \tag{3.8}$$

Efficiency, of Carnot's engine, is then, given by

$$\eta = \frac{W}{Q_1}$$
$$= \frac{nR\ (T_1 - T_2)\ ln\left(\frac{V_b}{V_a}\right)}{nR\ T_1\ ln\left(\frac{V_b}{V_a}\right)}$$

$$\boxed{\eta = 1 - \frac{T_2}{T_1}}$$

since $T_2 < T_1 \quad \Rightarrow \eta < 1$. From first definition of efficiency $\eta = 1 - \frac{Q_2}{Q_1}$, comparing with the previous result,

$$\boxed{\frac{Q_2}{Q_1} = \frac{T_2}{T_1}}$$

Note:

1. Efficiency of the Carnot's engine depends on the temperature of the source & sink and independent of the nature of the working substance.

2. If $T_2 = 0$, then $\eta = 1 =$ *maximum*. But the sink can not be maintained at absolute zero temperature. And therefore, 100% conversion of heat into work is not possible.

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3. If $T_1 = T_2$, thean $\eta = 0$ and the engine does not work,

4. The efficiency of a Carnot's engine can be increased by decreasing the ration T_2/T_1. This can be done either by decreasing T_2 or by increasing T_1. However, since $T_2 < T_1$, a decrease in T_2 is more effective than an equal increase in T_1.

5. A Carnot's engine is perfectly idealized engine because it is supposed to be free from alll losses due to conduction, radiation or friction. All the operations in Carnot's cycle are carried out quasi-statically and perfectly reversible.

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4 Second law of thermodynamics (direction of heat flow)

4.1 Kelvin-Planck's statement:

No process is possible whose sole result is the complete conversion of heat into work, i.e., one cannot devise a machine which just absorbs heat and produces 100% work.

4.2 Clausius' statement:

No process is possible whose sole result is the transfer of heat from a body at lower temperature to abody at higher temperature unaided by an external egency.

4.3 Carnot's theorem

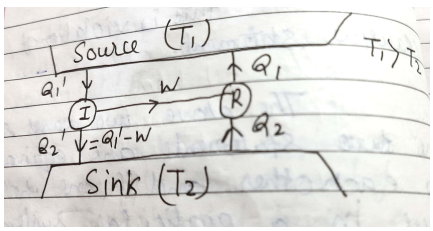
The kapild engines eye and are does form a self acting machine which unided by any external agency is transferring heat from a body at low temperature to a body at high temperature. This is invaluation of clausius statement of second law of thermodynamics implying their by that are initial assumption was incorrect

Part 1

No engine can be more efficient than a perfectly reversible engine working between the same two temperatures.

Part 2

All reversible engines working between the same two temperatures have the same efficiency.



Proof of first part: Let two engines I (irreversible) and R (reversible) operating between two heat reservoirs of temperature T_1 and T_2 such that $T_1 > T_2$. Let the amounts of working substance adjusted that the work perform by each is same. Let Q_2 be the amount of heat absorbed by R and Q_1 is amount of heat rejected by R. Q_1' is absorbed by I and Q_1' is rejected by I. η_R is the efficiency of R and eta_I is the efficiency of E.

$$W = Q_1 - Q_2 = Q_1' - Q_2' \tag{4.1}$$

<u>To prove: $\eta_R > \eta_I$</u> Let us consider that reverse holds true i.e., $\eta_I > \eta_R$. Couple the two engines and regard the system of a two combined engines to be a single device and engine

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I is driving the engine R backwards as a refrigerator. According to our assumption

$$\begin{aligned}
\eta_I &> \eta_R \\
\Rightarrow \frac{W}{Q_1'} &= \frac{W}{Q_1} \\
\Rightarrow Q_1 &> Q_1' \\
\because W = Q_1' - Q_2' &= Q_1 - Q_2 \\
\therefore Q_2 &> Q_2' \qquad (4.2)
\end{aligned}$$

Thus the net result of the coupled engine is as follows gain of heat by source $= Q_1 - Q_1'$ loss of heat by sink $= Q_2 - Q_2'$ external work done on the system I & $R = 0$. The coupled engines I and R thus form a self acting machine which unaided by any external agency is transferring heat from a body at low temperature to a body at high temperature. This is in voilation of clausius statement of second law of thermodynamics implying their by that are initial assumption was incorrect. therefore,

$$\boxed{\eta_R > \eta_I}$$

using this relation, we can prove that $\eta_{R_1} = \eta_{R_2}$.

Q: $\eta = 1/6$**, on decreasing the temperature of sink by** $65°$**C,** η **increases to** $1/3$**. Calculate the temperature of source and sink. [Ans:**$T_1 = 390°C$ **and** $T_2 = 325°C$**]**

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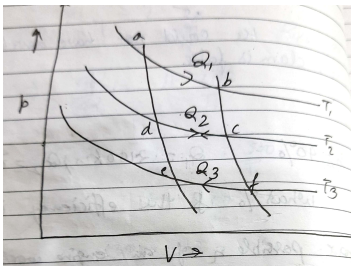
5 Entropy

Consider the diagram show in figure, two adiabats and three isotherms are shown. From Carnot's theorem,



$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$
$$\text{and } \frac{Q_2}{T_2} = \frac{Q_3}{T_3}$$

therefore

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} = \frac{Q_3}{T_3} = constant$$

this quantity was named as *entropy.*

$$\boxed{\frac{Q}{T} = S}$$

If adiabats are very closed to each other,

$$\frac{\delta Q}{T} = dS$$
$$\delta Q = T\ dS \tag{5.1}$$

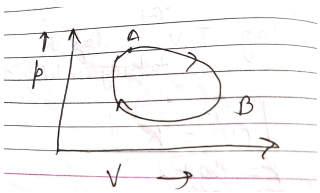
therefore, another form of first law of thermodynamics is given by

$$\boxed{TdS = C_V dT\ +\ pdV}$$

Change in entropy in an adiabatic process By the definition of entropy

$$dS = \frac{\delta Q}{T}$$
$$\because \delta Q = 0$$
$$\therefore dS = 0$$
$$\Rightarrow S = constant \tag{5.2}$$

Change in entropy in a cyclic process



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Consider a cyclic process as shown in the figure.
<u>from a $\rightarrow$ b</u> Gain in entropy

$$\int_a^b dS = \frac{Q_1}{T_1}$$

<u>from b $\rightarrow$ c</u> For an adiabatic process

$$\int_b^c dS = 0$$

<u>from c $\rightarrow$ d</u> Loss in entropy

$$\int_c^d dS = \frac{Q_2}{T_2}$$

<u>from d $\rightarrow$ a</u> adiabatic process

$$\int_d^a dS = 0$$

Net gain in entropy during $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$ i.e., cyclic process

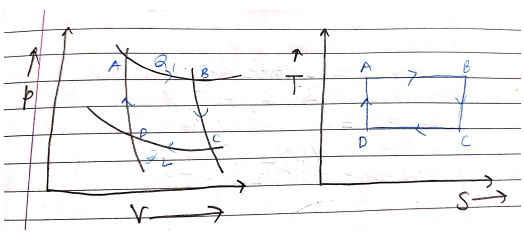
$$\oint dS = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} \tag{5.3}$$

Since for a carnot cycle $\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$, therefore

$$\oint dS = 0$$

Q: Calculate the change in entropy of 1 mol of a perfect gas which undergoes a change from p_1, V_1, T_1 to p_2, V_2, T_2. [Ans: $\Delta S = C_p \, ln\left(\frac{V_2}{V_1}\right) + C_V \, ln\left(\frac{p_2}{p_1}\right)$]

T-S diagram



5.1 Physical significance of entropy

Entropy is definite, single valued function of the thermodynamical variables describing the state of the working substance. In every natural process, there is an increase in entropy. Entropy is a measure of disorder, lack of information, it increases when heat is added to the system and decreases when heat is taken away from the system. A decrease in entropy implies a transition from chaotic or random state to a more ordered state. It is and extensive parameter whose units are Joule/Kelvin.

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5.2 Clausius inequality

let η' be the efficiency of an irreversible cyclic engine operating between T_1 and T_2. Let η be the efficiency of a revercible engine operating between the same temperatures. If Q_1 is the heat absorbed by E and Q_2 is the heat rejected by E, the by Carnot's theorem

$$\begin{aligned} \eta' &< \eta \\ \Rightarrow \frac{Q_1' - Q_2'}{Q_1'} &< \frac{Q_1 - Q_2}{Q_1} \\ \Rightarrow 1 - \frac{Q_2'}{Q_1'} &< 1 - \frac{Q_2}{Q_1} \\ \Rightarrow \frac{Q_2'}{Q_1'} &> \frac{Q_2}{Q_1} \end{aligned} \tag{5.4}$$

for a reversible engine

$$\begin{aligned} \frac{Q_2}{Q_1} &= \frac{T_2}{T_1} \\ \Rightarrow \frac{Q_2'}{T_2} &> \frac{Q_1'}{T_1} \\ \Rightarrow \oint \frac{\delta Q}{T} &= \frac{Q_1'}{T_1} - \frac{Q_2'}{T_2} \ < \ 0 \end{aligned}$$

Thus the value of the cyclic integral $\frac{\delta Q}{T}$ will be less than zero for an irreversible process. For a reversible process $\oint \frac{\delta Q}{T} = 0$, combining the two results, we get the Clausius inequality as

$$\boxed{\oint \frac{\delta Q}{T} <= 0}$$

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6 Maxwell's relations

Form first law of thermodynamics

$$dU = \delta Q - \delta W$$

since $\delta Q = TdS$ and $\delta W = pdV$

$$dU = TdS \ - \ pdV$$

Let $U,\ S$ & V be the function of independent variables x & y

$$\begin{aligned} U &\equiv U(x,y), \\ S &\equiv S(x,y) \ \& \\ V &\equiv V(x,y) \end{aligned} \tag{6.1}$$

the total derivative is, then given by

$$dU = \left(\frac{\partial U}{\partial x}\right)_y dx \ + \left(\frac{\partial U}{\partial y}\right)_x dy \tag{6.2}$$

$$dS = \left(\frac{\partial S}{\partial x}\right)_y dx \ + \ \left(\frac{\partial S}{\partial y}\right)_x dy$$

$$dU = \left(\frac{\partial V}{\partial x}\right)_y dx \ + \ \left(\frac{\partial V}{\partial y}\right)_x dy \tag{6.3}$$

substituting these values in (6.1) and comparing the coefficients of x & y

$$\left(\frac{\partial U}{\partial x}\right)_y = T\left(\frac{\partial S}{\partial x}\right)_y \ - \ p\left(\frac{\partial V}{\partial x}\right)_y \tag{6.4}$$

$$\left(\frac{\partial U}{\partial y}\right)_x = T\left(\frac{\partial S}{\partial y}\right)_x \ - \ p\left(\frac{\partial V}{\partial y}\right)_x \tag{6.5}$$

differentiating (6.4) with respect to y at constant x and differentiating (6.5) with respect to x at constant y and equating them, we get

$$\left(\frac{\partial p}{\partial x}\right)_y \left(\frac{\partial V}{\partial y}\right)_x \ - \ \left(\frac{\partial p}{\partial y}\right)_x \left(\frac{\partial V}{\partial x}\right)_y = \left(\frac{\partial T}{\partial x}\right)_y \left(\frac{\partial S}{\partial y}\right)_x \ - \ \left(\frac{\partial T}{\partial y}\right)_x \left(\frac{\partial S}{\partial x}\right)_y$$

The above relation is generic, to derive Maxwell's relations, we substitute in place of x&y any two of the four independent variables $S,\ p,\ T,\ V$.

first relation: Taking $x = T$ and $y = V$ in above equation,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

similarly, we can derive other three relations. Writing all four Maxwell's relations

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$$\boxed{\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V}$$

$$\boxed{\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p}$$

$$\boxed{\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p}$$

$$\boxed{\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V}$$

6.1 Application of Maxwell's relations

Specific heat equations

By definition of heat capacities

$$C_p = \left(\frac{\delta Q}{\delta T}\right)_p = T\left(\frac{\partial S}{\partial T}\right)_p$$
$$C_V = \left(\frac{\delta Q}{\delta T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V \tag{6.6}$$

Considering S as a function of T &V

$$dS = \left(\frac{\delta S}{\delta T}\right)_V dT + \left(\frac{\delta S}{\delta V}\right)_T dV$$
$$\Rightarrow T\left(\frac{\delta S}{\delta T}\right)_p = T\left(\frac{\delta S}{\delta T}\right)_V + T\left(\frac{\delta S}{\delta V}\right)_T \left(\frac{\delta V}{\delta T}\right)_p \tag{6.7}$$

from (6.6)

$$C_p - C_V = T\left(\frac{\delta S}{\delta V}\right)_T \left(\frac{\delta V}{\delta T}\right)_p$$

applying first Maxwell's equation $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$

$$\boxed{C_p - C_V = T\left(\frac{\delta p}{\delta T}\right)_V \left(\frac{\delta V}{\delta T}\right)_p}$$

Q: Use Maxwell's equation to derive $C_p - C_V = R$ for an ideal gas. Solution: Using equation of state for an ideal gas

$$p\,V = R\,T$$
$$\Rightarrow \left(\frac{\delta p}{\delta T}\right)_V = \frac{R}{V}$$
$$\& \left(\frac{\delta V}{\delta T}\right)_p = \frac{R}{p} \tag{6.8}$$

using heat equation $C_p - C_V = T\left(\frac{\delta p}{\delta T}\right)_V \left(\frac{\delta V}{\delta T}\right)_p$

$$\Rightarrow C_p - C_V = T\,\frac{R}{V}\,\frac{R}{p}$$
$$\Rightarrow C_p - C_V = R$$

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Q: Use specific heat equation to derive $C_p - C_V = R\left[1 + \frac{2a(V-b)^2}{RTV^3}\right]$ for a real gas.

Q:Show that $C_p - C_V = -T\ \left(\frac{\partial p}{\partial V}\right)_T\ \left(\frac{\partial V}{\partial T}\right)_p^2$

Hint: take $p \equiv p(T, V)$.

Q:Show that $C_p - C_V = T\ E_T\ \alpha_p^2\ V$.

Q:Show that $C_p - C_V = \frac{T\alpha^2\ V}{\beta_T}$, where $\beta_T = \frac{1}{E_T}$.

6.1.1 Clausius Clapyron equation (Change of phase)

Whenever there is a change of state from solid to liquid a liquid to Vapour the temperature remains constant as far as the change takes place. The clausius Clapyron equation relates the temperature that is melting point or boiling point with the pressure. From maxwell's relations

$$\begin{aligned}
\left(\frac{\partial S}{\partial V}\right)_T &= \left(\frac{\partial p}{\partial T}\right)_V \\
\Rightarrow T\ \left(\frac{\partial S}{\partial V}\right)_T &= \left(\frac{\partial p}{\partial T}\right)_V \\
\Rightarrow \left(\frac{\partial Q}{\partial V}\right)_T &= \left(\frac{\partial p}{\partial T}\right)_V
\end{aligned} \tag{6.9}$$

The quantity on lhs represents the quantity of heat absorbed or liberated per unit change in volume at constant temperature. As this change is taking place at constant temperature, the heat represents the latent heat. Let L be the amount of heat required to change the state of unit mass of the substance and let V_2 & V_1 be the corresponding volume. Then for small mass dm, we have

$$\begin{aligned}
\delta Q &= L\ dm \\
\&\ \delta V &= (V_2 - V_1)\ dm
\end{aligned} \tag{6.10}$$

therefore, from (6.9)

$$\frac{L\ dm}{(V_2 - V_1)\ dm} = T\ \frac{\partial p}{\partial T}$$

$$\boxed{\Rightarrow \frac{dp}{dT} = \frac{L}{T\ (V_2 - V_1)} = \frac{L}{T\ \Delta V}}$$

For evaporation process $V_2 = v_{vap}$ &$V_1 = V_{liquid}$

$$\begin{aligned}
\therefore \frac{dp}{dT} &= \frac{L}{T(V_{vap} - V_{liq.})} \\
\because v_{vap} &> V_{liq.} \\
\Rightarrow \frac{dp}{dT} &= +ve
\end{aligned} \tag{6.11}$$

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This means that pressure and temperature are related. An increase in vapour pressure raising the boiling point that is why food is cooked quickly in pressure cooker and why it is difficult to cook food at higher altitude. The clausius clapyron equation applies to all changes of phase, in which there is discontinuity in entropy and volume at the transition temperature. These are known as *first order phase transitions.*

Q: Show that the ratio of the adiabatic and isobaric coefficient of expansion is $\frac{\alpha_S}{\alpha_p} = \frac{1}{\gamma}$.

6.1.2 TdS Equations

Depending on the choice of independent variables using Maxwell's relation, we can obtain three TdS equations. 1st **TdS equation:** Let $S \equiv S(T, V)$

$$\begin{aligned}
\Rightarrow dS &= \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \\
\Rightarrow TdS &= C_V dT + T\left(\frac{\partial p}{\partial T}\right)_V dV \qquad (6.12) \\
\because E_T &= -V\left(\frac{\partial p}{\partial V}\right)_T \\
\alpha &= \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p \\
\Rightarrow E_T &= -\left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \\
&= \left(\frac{\partial p}{\partial T}\right)_V
\end{aligned}$$

$$\boxed{TdS = C_V\ dT + TE_T\alpha dV}$$

2nd **TdS equation:**

$$\begin{aligned}
S &\equiv S(T, p) \\
\Rightarrow TdS &= C_p\ dT - TV\alpha dp \qquad (6.13)
\end{aligned}$$

3rd **TdS equation:**

$$\begin{aligned}
S &\equiv S(p, V) \\
\Rightarrow TdS &= C_V\left(\frac{\partial T}{\partial p}\right)_V dp + C_p\left(\frac{\partial T}{\partial V}\right)_p dV \qquad (6.14)
\end{aligned}$$

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6.1.3 Energy equations

1st **energy equation:**

$$
\begin{aligned}
U &\equiv U(T,V) \\
\therefore dU &= \left(\frac{\partial U}{\partial T}\right)_V dT \; + \; \left(\frac{\partial U}{\partial V}\right)_T dV \\
\because dU &= TdS \; - \; pdV \\
\Rightarrow \left(\frac{\partial U}{\partial V}\right)_T &= T \; \left(\frac{\partial S}{\partial V}\right)_T \; - \; p
\end{aligned}
\tag{6.15}
$$

$$
\boxed{\left(\tfrac{\partial U}{\partial V}\right)_T \; = \; T \; \left(\tfrac{\partial p}{\partial T}\right)_V \; - \; p}
$$

Q: Calculate $\left(\frac{\partial U}{\partial V}\right)_T$ for ideal gas and real gas. Solution:

<u>For an ideal gas</u> $pV = RT$

$$
\begin{aligned}
p &= \frac{RT}{V} \\
\Rightarrow \left(\frac{\partial p}{\partial T}\right)_V &= \frac{R}{V}
\end{aligned}
\tag{6.16}
$$

substituting this in first energy equation,

$$
\begin{aligned}
\left(\frac{\partial U}{\partial V}\right)_T &= \frac{RT}{V} - p \\
&= p - p \\
\Rightarrow \left(\frac{\partial U}{\partial V}\right)_T &= 0
\end{aligned}
\tag{6.17}
$$

it means that the internal energy of the ideal gas is dependent only on the kinetic energy and independent of volume.

<u>For a real gas</u> $\left(p + \frac{a}{V^2}\right)(V - b) = RT$

$$
\begin{aligned}
\Rightarrow \left(\frac{\partial p}{\partial T}\right)_V &= \frac{R}{V-b} \\
\therefore \left(\frac{\partial U}{\partial V}\right)_T &= \frac{RT}{V-b} - p \\
\Rightarrow\Rightarrow \left(\frac{\partial U}{\partial V}\right)_T &= \frac{a}{V^2}
\end{aligned}
\tag{6.18}
$$

Q: Show that $dU \; = \; C_V dT \; + \; \left[T\left(\frac{\partial p}{\partial T}\right)_V \; - \; p\right] \; dV$**.**

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1st **energy equation:**

$$\because dU = \delta Q \ - pdV$$
$$\therefore \left(\frac{\partial U}{\partial p}\right)_T = = T\left(\frac{\partial S}{\partial p}\right)_T - \ p\left(\frac{\partial V}{\partial p}\right)_T$$
$$\therefore \left(\frac{\partial U}{\partial p}\right)_T = = -T\left(\frac{\partial V}{\partial T}\right)_p - \ p\left(\frac{\partial V}{\partial p}\right)_T$$

6.1.4 Temperature change during adiabatic process

From Maxwell's equation

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$
$$\Rightarrow \left(\frac{\partial T}{\partial V}\right)_S = -T\ \left(\frac{\delta p}{\delta Q}\right)_V$$

An increase in amount of heat δQ at constant Volume results in an increase in pressure, thus

$$\frac{\delta p}{\delta Q} = +ve$$
$$\Rightarrow \frac{\partial T}{\partial V} = -ve$$

therefore, during an adiabatic process T decreases when V increases. Also

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$
$$= T\ \left(\frac{\delta V}{\delta Q}\right)_p$$

because, when heat supply increases, then volume increases, therefore

$$\frac{\delta V}{\delta Q} = +ve$$
$$\therefore \left(\frac{\partial T}{\partial p}\right)_S = +ve \tag{6.19}$$

thus when p increaes, the temperature T increases.

6.1.5 Adiabatic lapse rate

It is defined as the variation of temperature in the atmosphere with height above the sea level. The main cause of this lapse rate is due to the Convection currents in the troposphere, which is the lowest region of the earth's atmosphere. The Convection currents continuously transport hot air from the lower regions to the higher ones and vice versa. When the air from the sea level rises to the upper region of lower pressure, it expands, since air is a poor conductor of heat, very little heat is exchange during this expansion. Hence the expansion maybe considered adiabatic.

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Pressure of air at height h = p.

$$\because p = h\ \rho g$$
$$\Rightarrow dp = -\rho\ g\ dh \quad \text{as height increases, p decreases}$$

If the gas of consideration is an ideal gas,

$$p\ V = R\ T$$
$$\therefore \rho = \frac{M}{\frac{RT}{p}}$$
$$\Rightarrow \rho = \frac{Mp}{RT}$$
$$\Rightarrow \frac{dp}{p} = -\frac{Mg}{RT}\ dh \tag{6.20}$$

integrating on both the sides,

$$ln(p) = -\frac{Mgh}{RT}\ +\ \kappa$$
$$\text{At} \quad h = 0,\ \ p = p_0 \quad \Rightarrow \kappa = ln(p_0)$$
$$\Rightarrow p = p_0\ e^{-\frac{Mgh}{RT}} \tag{6.21}$$

defining adiabatic lapse rate $alpha$ as $-\frac{dT}{dh}$, and at $h = 0$ $T = T_0$,

$$\therefore \alpha = -\frac{T_0 - T}{0 - h}$$
$$\Rightarrow \alpha h = T_0 - T$$
$$\Rightarrow T = T_0\ -\ \alpha\ h \tag{6.22}$$

$\because \frac{dp}{p} = -\frac{Mgdh}{RT}$, substituting $T = T_0 - \alpha h$, we have

$$\frac{dp}{p} = -\frac{Mgdh}{R(T_0 - \alpha h)}$$

integrating on both the sides,

$$p = p_0\ e^{\frac{Mg}{\alpha R} ln\left(\frac{T_0 - \alpha h}{T_0}\right)} \tag{6.23}$$

For an adiabatic process $p^{\gamma-1}\ T^{-gamma} = constant$, differentiating partially Poisson's equation,

$$\frac{dp}{p} = \frac{\gamma}{\gamma - 1}\frac{dT}{T}$$

also

$$\frac{dp}{p} = -\frac{Mg}{RT}\ dh$$

comparing both the relations, we get

$$\boxed{\frac{dT}{dh} = \frac{M\ g}{R}\ \left(\frac{1\ -\ \gamma}{\gamma}\right)}$$

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